

Measures of Center and Spread

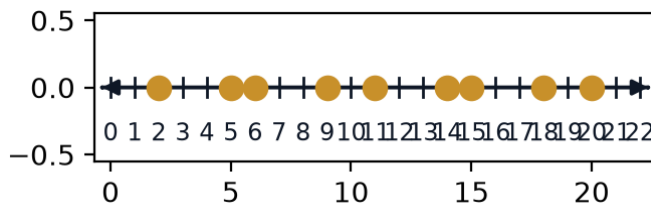


Mean and median

- 1 Find the mean and median of the data set: 4, 7, 7, 9, 13.
- 2 A sixth data point of 40 is added to the set 4, 7, 7, 9, 13. Recompute the mean and median, and explain which measure changed more.

Range, IQR, and quartiles

- 3 Find the five-number summary (min, Q_1 , median, Q_3 , max) of the ordered data: 2, 5, 6, 9, 11, 14, 15, 18, 20.



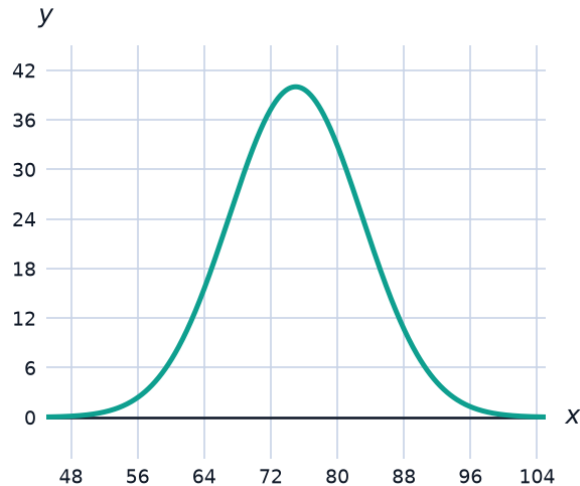
- 4 Using the five-number summary from the previous problem, find the IQR and determine whether 20 would be flagged as an outlier using the $1.5 \times \text{IQR}$ rule.

Standard deviation and variance

- 5 Find the standard deviation of the data set 2, 4, 6, 8 (treat as a population).
- 6 Explain why the standard deviation is always non-negative, and why it equals 0 only in one specific case.

Visualizing spread with a normal-shaped distribution

- 7 The graph shows a bell-shaped distribution of exam scores with mean 75 and standard deviation 8.



- a Estimate the range of scores that fall within one standard deviation of the mean.
- b Estimate the range of scores that fall within two standard deviations of the mean.

Comparing distributions using measures of spread

- 8 Class A has a mean test score of 80 with a standard deviation of 4. Class B has a mean of 80 with a standard deviation of 12. Compare the two classes' consistency, and explain what a small vs. large standard deviation indicates.

Resistant vs. non-resistant statistics

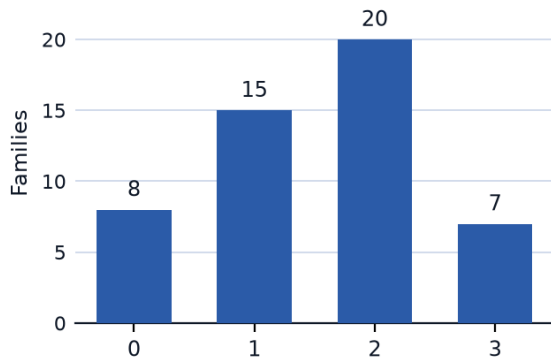
- 9 Explain why the median and IQR are described as "resistant" measures, while the mean and standard deviation are not, using the outlier example from earlier in this worksheet as a reference.

Choosing appropriate summary statistics

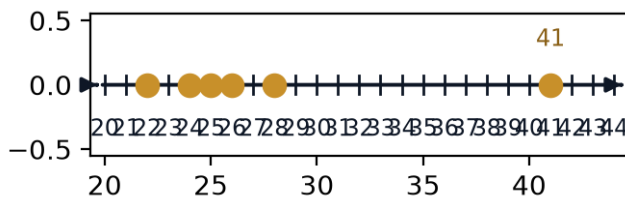
- 10 A data set of home prices in a neighborhood is strongly skewed right due to a few very expensive mansions. Recommend which pair of summary statistics (mean/standard deviation or median/IQR) should be reported, and justify your choice.
- 11 A quality-control process measures the diameter of manufactured bolts, which are expected to be very consistent (low variability) and roughly symmetric. Recommend which pair of summary statistics is more appropriate here, and explain why.

Computing summary statistics from a frequency table

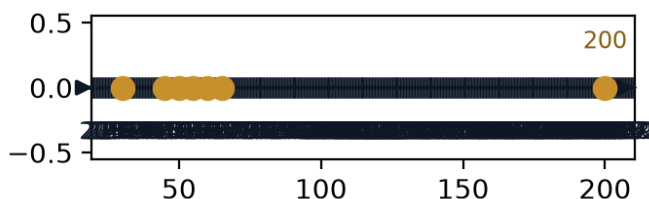
- 12** A survey of families records number of children per household: 0 children (8 families), 1 child (15 families), 2 children (20 families), 3 children (7 families). Find the mean number of children per household.



- 13** A small business records weekly sales (in thousands): 10, 12, 12, 14, 17. Find the mean, and then find the variance and standard deviation (treat as a population).
- 14** Two data sets both represent quiz scores out of 20 for two different classes of the same size, and both have the same mean of 15. Class A has a standard deviation of 1.2, while Class B has a standard deviation of 4.5. Without doing any further calculation, describe what a dotplot of each class's scores would likely look like, and explain how you can tell just from the two standard deviations.
- 15** A coach records the vertical jump heights (in inches) of 7 athletes: 22, 24, 25, 26, 26, 28, 41. Explain why the value 41 would likely be flagged as an outlier, and describe, in general terms, the two-step process for using the $1.5 \times \text{IQR}$ rule to formally check whether a value is an outlier.



- 16** A teacher records the number of minutes 8 students spent on homework last night: 30, 45, 50, 55, 60, 60, 65, 200. The value 200 comes from a student who mistakenly reported total weekly homework time instead of just last night. Explain, step by step, how this single incorrect data entry would affect the reported mean versus the reported median, and explain which measure a teacher should trust more when summarizing typical homework time for this data set, given the known data-entry error.



Multiple choice — easy 15

17 Find the mean of the data set: 17, 28, 12, 6, 13.

a. $\frac{81}{5}$

b. $\frac{71}{5}$

c. 19

d. $\frac{76}{5}$

18 Find the mean of the data set: 21, 17, 4, 14, 29.

a. 16

b. 17

c. 18

d. $\frac{85}{4}$

19 Find the mean of the data set: 1, 8, 15, 8, 7.

a. $\frac{39}{5}$

b. $\frac{44}{5}$

c. $\frac{39}{4}$

d. $\frac{34}{5}$

20 Find the mean of the data set: 4, 19, 10, 17, 14.

a. $\frac{64}{5}$

b. $\frac{69}{5}$

c. 16

d. $\frac{59}{5}$

21 Find the mean of the data set: 15, 26, 21, 10, 14.

a. $\frac{43}{2}$

b. $\frac{91}{5}$

c. $\frac{86}{5}$

d. $\frac{81}{5}$

22 Find the mean of the data set: 15, 11, 5, 7, 11.

a. $\frac{49}{4}$

b. $\frac{54}{5}$

c. $\frac{49}{5}$

d. $\frac{44}{5}$

23 Find the mean of the data set: 15, 23, 24, 25, 9.

a. $\frac{101}{5}$

b. $\frac{91}{5}$

c. $\frac{96}{5}$

d. 24

24 Find the mean of the data set: 12, 1, 29, 2, 16.

a. 12

b. 11

c. 13

d. 15

25 Find the mean of the data set: 3, 28, 18, 14, 29.

a. 23

b. $\frac{97}{5}$

c. $\frac{87}{5}$

d. $\frac{92}{5}$

26 Find the mean of the data set: 22, 24, 23, 15, 27.

a. $\frac{106}{5}$

b. $\frac{111}{4}$

c. $\frac{111}{5}$

d. $\frac{116}{5}$

27 Find the mean of the data set: 4, 8, 8, 7, 6.

a. $\frac{33}{4}$

b. $\frac{33}{5}$

c. $\frac{38}{5}$

d. $\frac{28}{5}$

28 Find the mean of the data set: 26, 7, 1, 1, 20.

a. 11

b. 12

c. $\frac{55}{4}$

d. 10

29 Find the mean of the data set: 19, 15, 3, 8, 28.

a. $\frac{73}{5}$

b. $\frac{73}{4}$

c. $\frac{78}{5}$

d. $\frac{68}{5}$

30 Find the mean of the data set: 12, 15, 25, 7, 25.

a. $\frac{89}{5}$

b. 21

c. $\frac{84}{5}$

d. $\frac{79}{5}$

31 Find the mean of the data set: 14, 16, 29, 17, 21.

a. $\frac{92}{5}$

b. $\frac{97}{5}$

c. $\frac{97}{4}$

d. $\frac{102}{5}$

Multiple choice — medium 15

32 Find the median of the data set: 16, 15, 5, 25, 29, 12.

a. $\frac{31}{2}$

b. 17

c. $\frac{33}{2}$

d. $\frac{37}{2}$

33 Find the median of the data set: 8, 37, 13, 24, 10, 25.

a. $\frac{35}{2}$

b. 18

c. $\frac{37}{2}$

d. $\frac{39}{2}$

34 Find the median of the data set: 36, 31, 25, 26, 21, 40.

a. $\frac{61}{2}$

b. $\frac{59}{2}$

c. $\frac{179}{6}$

d. $\frac{57}{2}$

35 Find the median of the data set: 31, 29, 29, 23, 23, 36.

a. 30

b. 27

c. $\frac{57}{2}$

d. 29

36 Find the median of the data set: 8, 19, 2, 26, 32, 35.

a. $\frac{45}{2}$

b. 20

c. $\frac{47}{2}$

d. $\frac{61}{3}$

37 Find the median of the data set: 12, 30, 16, 1, 22, 37.

a. 21

b. 19

c. $\frac{59}{3}$

d. 20

38 Find the median of the data set: 16, 25, 21, 39, 37, 32.

a. $\frac{85}{3}$

b. 29

c. $\frac{59}{2}$

d. $\frac{57}{2}$

39 Find the median of the data set: 24, 30, 40, 6, 31, 5.

a. $\frac{37}{2}$

b. 27

c. $\frac{68}{3}$

d. 28

40 Find the median of the data set: 1, 39, 36, 20, 23, 10.

a. $\frac{45}{2}$

b. $\frac{43}{2}$

c. 23

d. 21

41 Find the median of the data set: 16, 31, 25, 32, 7, 37.

a. $\frac{74}{3}$

b. 29

c. 28

d. 24

42 Find the median of the data set: 25, 32, 13, 21, 40, 31.

a. $\frac{53}{2}$

b. 29

c. 27

d. 28

43 Find the median of the data set: 6, 37, 23, 38, 20, 2.

a. 21

b. $\frac{41}{2}$

c. $\frac{43}{2}$

d. $\frac{45}{2}$

44 Find the median of the data set: 10, 13, 40, 18, 9, 14.

a. $\frac{52}{3}$

b. 14

c. $\frac{27}{2}$

d. $\frac{29}{2}$

45 Find the median of the data set: 24, 32, 5, 37, 2, 37.

- a. 29
- b. 21
- c. 28
- d. $\frac{137}{6}$

46 Find the median of the data set: 22, 12, 30, 18, 11, 28.

- a. $\frac{121}{6}$
- b. $\frac{39}{2}$
- c. 21
- d. 20

Multiple choice — hard 15

47 Find the IQR of the ordered data set: 4, 7, 20, 29, 43, 50, 51, 54.

- a. 36
- b. $\frac{101}{2}$
- c. 38
- d. 37

48 Find the IQR of the ordered data set: 2, 8, 10, 12, 20, 23, 34, 45.

- a. $\frac{39}{2}$
- b. $\frac{41}{2}$
- c. $\frac{37}{2}$
- d. $\frac{57}{2}$

- 49 Find the IQR of the ordered data set: 2, 5, 6, 13, 23, 29, 34, 53.
- a. 26
 - b. $\frac{63}{2}$
 - c. 25
 - d. 27
- 50 Find the IQR of the ordered data set: 6, 10, 14, 16, 22, 23, 34, 50.
- a. $\frac{57}{2}$
 - b. $\frac{33}{2}$
 - c. $\frac{35}{2}$
 - d. $\frac{31}{2}$
- 51 Find the IQR of the ordered data set: 18, 21, 35, 38, 39, 46, 55, 58.
- a. $\frac{43}{2}$
 - b. $\frac{47}{2}$
 - c. $\frac{45}{2}$
 - d. $\frac{101}{2}$
- 52 A data set has $Q_1 = 22$ and $Q_3 = 43$. Using the $1.5 \times \text{IQR}$ rule, find the upper fence used to detect outliers.
- a. $\frac{147}{2}$
 - b. 64
 - c. $\frac{149}{2}$
 - d. $\frac{151}{2}$

- 53 A data set has $Q_1 = 24$ and $Q_3 = 46$. Using the $1.5 \times \text{IQR}$ rule, find the upper fence used to detect outliers.
- a. 78
 - b. 68
 - c. 79
 - d. 80
- 54 A data set has $Q_1 = 25$ and $Q_3 = 32$. Using the $1.5 \times \text{IQR}$ rule, find the upper fence used to detect outliers.
- a. $\frac{85}{2}$
 - b. $\frac{83}{2}$
 - c. $\frac{87}{2}$
 - d. 39
- 55 A data set has $Q_1 = 26$ and $Q_3 = 48$. Using the $1.5 \times \text{IQR}$ rule, find the upper fence used to detect outliers.
- a. 80
 - b. 70
 - c. 81
 - d. 82
- 56 A data set has $Q_1 = 14$ and $Q_3 = 28$. Using the $1.5 \times \text{IQR}$ rule, find the upper fence used to detect outliers.
- a. 48
 - b. 50
 - c. 42
 - d. 49

- 57 Find the standard deviation of the data set 14, 10, 13, 23 (treat as a population), to two decimal places.
- a. 5.15
 - b. 4.85
 - c. 4.55
 - d. 5.45
- 58 Find the standard deviation of the data set 29, 22, 31, 38 (treat as a population), to two decimal places.
- a. 6.30
 - b. 5.40
 - c. 5.70
 - d. 6.00
- 59 Find the standard deviation of the data set 15, 7, 15, 3 (treat as a population), to two decimal places.
- a. 4.90
 - b. 5.20
 - c. 5.50
 - d. 5.80
- 60 Find the standard deviation of the data set 22, 20, 16, 18 (treat as a population), to two decimal places.
- a. 2.54
 - b. 2.24
 - c. 2.84
 - d. 1.94

- 61 Find the standard deviation of the data set 35, 31, 24, 26 (treat as a population), to two decimal places.
- a. 4.90
 - b. 4.30
 - c. 4.60
 - d. 4.00